

Algorithms and Programs

TC1037 · Algorithmic Thinking · Lecture 01

Dr. Leopoldo Rodríguez Salazar

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Tecnológico de Monterrey · Campus Querétaro · Group 510

Tell me how to draw a square.

I will follow your instructions **exactly**.

What just happened

- Every instruction you gave me was **correct**.
- I produced garbage.
- I did not misunderstand you — I understood you **too literally**.

What was missing was never the *idea*.
It was precision about the things you thought were obvious.

A **finite, ordered, unambiguous** sequence of steps that turns a **defined input** into a **defined output**.

Algoritmo

The four properties

Finite It ends. “Keep adding numbers” is not an algorithm. “...until the user types 0” is.

Ordered Break the eggs *then* heat the pan $\neq$ heat the pan *then* break the eggs.

Unambiguous Exactly one reading. If two people get different answers, it is a bug you have not found.

Defined I/O You know what you are given and what you must produce.

Algorithm vs. Program

Algorithm

The **thinking**

Natural language, pseudocode, a
flowchart

Read by a human

Program

The **typing**

Python

Read by a machine

A program can fail for every reason an algorithm can — **plus syntax**.

Our notation

```
ALGORITHM   average_speed
INPUT:      distance (km), time (hours)
OUTPUT:     speed (km/h)

1. READ distance
2. READ time
3. speed <- distance / time
4. DISPLAY speed
END
```

- <- is assignment — *“gets the value of”*. A command, not a claim.
- Steps are numbered. That is what makes “ordered” checkable.

What happens if the user types 0 for the time?

How to write one that survives

1. State **input** and **output**. Two lines, before anything else.
2. Write the steps, **numbered**. One action per step.
3. **Attack it**. Read it as literally as I read your square.
4. Fix, and repeat.

Step 3 is the one that separates a passing grade from a good engineer.
Everyone can do step 2.

Exercise 1 — Price validation

Individual · on paper · 10 minutes

Write an algorithm that checks whether a price entered by the user is valid.
A price is valid if it is positive.

Then attack it, in writing:

- What does yours do if the user enters exactly 0?
- What if the user enters `hola`?
- Does it say what “valid” *looks like* as output?

Exercise 2 — Metres to feet

Pairs · on paper · 15 minutes

Given a length in metres, calculate and display the equivalent in feet.

1 foot = 12 inches 1 inch = 2.54 cm 1 m = 100 cm

1. Numbered steps, explicit INPUT: / OUTPUT:.
2. **Swap with the pair next to you.** Execute *their* algorithm literally for 3.5 metres.
3. Different answers? Find the step where you diverged. **Write down the word that caused it.**

16:30 – 17:30 Laboratory — this room, with Luis

17:35 – 18:55 Session 2 — me

Bring your laptop, a notebook and a pencil.

The lab opens with a 20-minute diagnostic that needs the laptop.

Everything after it is on paper.

Bring today's algorithm too — you will be attacking each other's.

Eat before you come. Not in the room.

Before Friday

1. Install **Thonny**. Bring it working — we write our first program Friday.
2. Install **Respondus LockDown Browser**; practice exam by Mon 28 Sep, 23:59.
3. Levelling module: units 1 (memory) and 2 (arithmetic expressions).
4. Read the two handouts: course policies, and *Using AI in this course*.

Only have a Mac? Tell me Friday. Macs are not permitted for exams.

**An algorithm is a specification
a hostile reader cannot misread.**